

PIEDMONT TRIAD, NC, USA

WMO: 723170

Lat: 36.097N

Lon: 79.943W

Elev: 890

StdP: 14.23

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13723

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.6	22.2	-1.0	5.4	24.6	3.5	6.8	28.8	24.8	49.1	22.0	48.0	6.9	300	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	92.4	74.0	90.3	73.5	88.2	72.9	77.0	87.2	75.8	85.3	75.0	83.9	7.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.1	131.7	80.4	73.1	127.2	79.5	72.2	123.3	78.8	40.7	87.3	39.8	85.7	38.9	84.0	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.4	(b) 18.0	(c) 16.0	(e) 11.4	(f) 95.7	(g) 5.1	(h) 2.5	(i) 7.6	(j) 97.5	(k) 4.6	(l) 99.0	(m) 1.7	(n) 100.3	(o) -2.0	(p) 102.1		
(a) 20.4	(b) 18.0	(c) 16.0	(e) 9.3	(f) 78.3	(g) 4.5	(h) 1.4	(i) 6.0	(j) 79.3	(k) 3.3	(l) 80.1	(m) 0.8	(n) 80.9	(o) -2.5	(p) 81.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.6	39.4	42.9	50.2	59.5	67.5	75.2	78.3	77.1	70.9	60.3	49.7	42.8		
	DBStd	15.66	10.15	9.29	9.48	8.10	6.84	5.18	4.01	4.36	5.94	7.61	8.37	8.97		
	HDD50	1102	356	234	118	15	0	0	0	0	0	10	107	262		
	HDD65	3495	795	619	466	200	58	4	0	0	18	185	460	690		
	CDD50	4594	26	36	123	301	542	757	878	839	627	328	99	38		
	CDD65	1513	0	1	6	36	135	311	414	374	195	38	2	1		
	CDH74	12749	0	3	53	341	1063	2676	3745	3221	1348	280	17	2		
	CDH80	4579	0	0	3	58	280	1008	1538	1226	412	53	1	0		
	WSAvg	7.0	7.6	7.9	8.5	8.7	7.4	6.6	6.1	5.7	6.1	6.2	6.5	6.8		
	PrecAvg	42.7	3.3	3.0	3.8	3.3	3.7	3.8	4.3	4.1	4.1	3.2	3.0	3.2		
(15) Precipitation	PrecMax	64.2	7.7	5.8	8.7	8.0	8.7	10.5	12.7	9.6	13.1	12.6	8.3	7.0		
	PrecMin	29.7	0.7	0.5	0.7	0.4	0.6	0.0	1.1	0.7	0.0	0.0	0.2	0.6		
	PrecStd	7.3	1.6	1.3	1.7	1.7	1.9	2.3	2.5	2.2	3.0	2.4	2.0	1.6		
	PrecStd	7.3	1.6	1.3	1.7	1.7	1.9	2.3	2.5	2.2	3.0	2.4	2.0	1.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.1	73.1	79.5	85.4	89.1	94.2	95.2	95.0	91.2	85.4	76.4	71.6		
		MCWB	58.8	60.4	60.9	65.0	70.6	73.0	75.3	74.5	71.2	71.4	62.7	61.0		
	2%	DB	64.1	67.3	74.7	81.3	85.9	91.0	92.5	91.8	87.7	80.6	71.4	65.9		
		MCWB	57.6	56.5	58.8	63.8	69.5	72.8	74.8	74.3	71.6	67.1	59.8	59.5		
	5%	DB	60.2	63.3	70.5	78.2	83.3	88.4	90.2	89.2	84.9	76.9	68.0	61.8		
		MCWB	53.9	54.5	56.7	62.3	68.4	72.3	74.3	73.7	70.7	64.9	59.4	56.1		
	10%	DB	55.5	58.5	65.9	74.5	80.2	85.9	87.9	86.6	81.8	73.3	64.3	57.6		
		MCWB	49.2	49.4	54.5	60.6	66.8	71.2	73.6	73.0	69.5	64.1	57.0	51.5		
	0.4%	WB	63.3	63.8	64.4	69.4	73.6	76.4	78.3	78.3	76.1	74.3	67.7	65.4		
		MCDB	65.8	68.5	73.2	77.4	83.2	87.2	90.0	88.5	84.6	79.4	70.3	68.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	59.4	60.0	62.0	66.9	71.7	75.1	77.1	77.0	74.5	71.1	64.4	61.2		
		MCDB	63.2	65.3	69.8	76.6	82.0	85.7	88.1	86.6	82.0	76.3	68.2	64.6		
	5%	WB	55.2	56.0	59.7	64.8	70.3	74.1	76.1	75.7	73.1	68.4	61.5	57.5		
		MCDB	58.9	61.3	67.2	74.1	79.6	84.3	86.1	84.4	79.6	74.0	66.1	60.9		
	10%	WB	50.2	51.2	57.1	62.9	68.9	73.0	75.1	74.7	71.7	65.6	58.0	52.6		
		MCDB	54.7	56.3	64.1	71.3	77.1	82.6	84.3	82.8	77.7	71.6	63.2	56.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	18.9	20.2	21.1	19.4	18.1	17.4	17.1	17.3	19.6	19.9	17.2		
		MCDBR	22.2	24.6	24.8	25.0	22.2	20.8	20.2	19.9	21.0	22.8	24.0	21.3		
		MCWBR	15.7	16.4	13.4	11.4	9.2	7.4	6.6	6.5	7.9	11.1	14.7	15.5		
		MCDBR	19.6	21.5	21.1	20.9	18.9	18.5	18.0	17.0	16.6	17.6	19.0	18.1		
	5% WB	MCWBR	16.5	17.1	13.3	11.0	8.6	7.4	6.6	6.5	7.3	10.7	15.1	15.0		
		MCWBR	16.5	17.1	13.3	11.0	8.6	7.4	6.6	6.5	7.3	10.7	15.1	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.304	0.318	0.349	0.391	0.434	0.473	0.515	0.506	0.410	0.368	0.320	0.306		
	taud		2.513	2.468	2.422	2.306	2.216	2.144	2.050	2.057	2.323	2.420	2.513	2.540		
	Ebn at Noon		285	293	292	283	271	259	247	247	268	271	276	277		
	Edn at Noon		25	30	34	41	45	49	53	52	38	31	25	23		
	RadAvg		803	1014	1348	1686	1865	2005	1870	1709	1427	1166	886	687		
(44) All-Sky Solar Radiation	RadStd		53	123	91	132	153	147	123	86	131	148	70	60		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.10	N/A	N/A	N/A	N/A	N/A	N/A	+326	+209
(47) Regional (62 neighbors)		+0.77	N/A	N/A	N/A	N/A	N/A	N/A	+252	+173

Nomenclature: See separate page